Modern Cryptography

Discrete logarithms and RSA

Mathematical Foundations

- Definition of the Multiplicative Group
- Fermat’s Little Theorem
- Cyclic Groups and Generators

Definition of the Multiplicative Group

Mathematical Foundations

- Definition of the Multiplicative Group
  - Definition: Muliplicative Group Z_p^*
    - set of integers moudulo prime p, excluding 0.
    - Z_p^* = {1, 2, ..., p-1}
  - Properties of Z_p^*:
    - Identity element: $id = 1$
    - Order of the group: $|Z_p^*| = p - 1$

Mathematical Foundations

- Definition of the Multiplicative Group
  - Fact (General Group Theory):
    - Let G be a group of order m and $a \in G$. Then, $a^m = id$
  - Application to Z_p^*:
    - Substitute $m = p - 1$ and id = 1

Fermat's Little Theorem

Fermat's Little Theorem

- Theorem:
  - p가 소수이고 a가 p로 나누어떨어지지 않는 양의 정수이면(즉, $gcd(a,p) = 1$), $a^{p-1} \equiv 1 \pmod{p}$이 항상 성립한다
- Proof:
  - $1a \times 2a \times 3a \times \cdots \times (p-1)a \equiv [1 \times 2 \times \cdots \times (p-1)] \pmod{p}$
  - $a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}$
  - (p-1)과 p는 서로소
  - $\therefore a^{p-1} \equiv 1 \pmod{p}$

Fermat's Little Theorem

- 설정: 소수 $p = 5$, 양의 정수 $a = 3$(gcd(3, 5) = 1)

  i. Multiplicative Group(Z_5^*)정의:

  - Z_5^* = {1, 2, 3, 4}
  - Order $m = p - 1 = 4$.

  ii. 좌변 전개 (집합의 모든 원소에 a = 3을 곱함):

  - $1 \times 3 = 3 \equiv 3 \pmod 5$
  - $2 \times 3 = 6 = (5 \times 1) + 1 \equiv 1 \pmod 5$
  - $3 \times 3 = 9 = (5 \times 1) + 4 \equiv 4 \pmod 5$
  - $4 \times 3 = 12 = (5 \times 2) + 2 \equiv 2 \pmod 5$
  - 결과 집합은 {3, 1, 4, 2}로, 순서만 바뀌었을 뿐 원래 집합과 동일하다.

Fermat's Little Theorem

- $(1 \times 3) \times (2 \times 3) \times (3 \times 3) \times (4 \times 3) = 3^4 \times (1 \times 2 \times 3 \times 4) = 3^4 \times 4!$
- 합동식: $3^4 \times 24 \equiv 24 \pmod 5$
- 24와 소수 5는 서로소($\gcd(24, 5) = 1$)이므로 양변을 24로 약분합니다.
- 결론: $3^4 \equiv 1 \pmod 5$ 가 도출됩니다.

- 지수화 검증:
  - 실제 $3^4 = 81$ 이며, $81 = (5 \times 16) + 1 \equiv 1 \pmod 5$ 로 증명과 일치한다.

Cyclic Groups and Generators

Cyclic Groups and Generators

- Definition: Generated Subgroup $\langle g \rangle$
  - Let G be a group of order m and let $g \in G$.
  - $\langle g \rangle = \{ g^i : i \in Z_m \}$.
- Fact (Order of an element):
  - The size $|\langle g \rangle|$ is defined to be the order of the element g.
  - $|\langle g \rangle|$ is always a divisor of the group order m.

Cyclic Groups and Generators

- Definition: Generaor (Primitive Element)
  - An element $g \in G$ is a generator of G if $|\langle g \rangle| = m$.
  - In this case, $\langle g \rangle = G$.
- Definition: Cyclic Group
  - We say that G is cyclic if it has at least one gernator.
- Fact (Cryptographic Application):
  - if p is a prime, the multiplication group Z_p^* is always cyclic.

Cyclic Groups and Generators

- Example of Generators in Z_{11}^*
  - Setting:
    - Group $G = Z_{11}^*$ = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}.
    - Order of group m = 10.
  - Case 1: $g = 5$ (Not a generator)
    - $5^1 \equiv 5, 5^2 \equiv 3, 5^3 \equiv 4, 5^4 \equiv 9, 5^5 \equiv 1 \pmod{11}$.
    - $\langle 5 \rangle$ = {1, 3, 4, 5, 9}, Order = $5 \neq 10$.
  - Case 2: $g = 2$ (Generator)
    - $2^1 \equiv 2, 2^2 \equiv 4, 2^3 \equiv 8, \ldots, 2^{10} \equiv 1 \pmod{11}$.
    - $\langle 2 \rangle = Z_{11}^*$, Order = 10.

Cyclic Groups and Generators

- Setting: 곱셈군 Z_{11}^*, 군의 위수 $m = 10$.
  - Case 1: $g = 5$, 원소 5를 거듭제곱하여 모듈러 11 연산을 수행.
    - $5^1 \equiv 5 \pmod{11}$
    - $5^2 = 25 = (11 \times 2) + 3 \equiv 3 \pmod{11}$
    - $5^3 = 5^2 \times 5 \equiv 3 \times 5 = 15 \equiv 4 \pmod{11}$
    - $5^4 = 5^3 \times 5 \equiv 4 \times 5 = 20 \equiv 9 \pmod{11}$
    - $5^5 = 5^4 \times 5 \equiv 9 \times 5 = 45 \equiv 1 \pmod{11}$
  - 결론: $5^5 \equiv 1 \pmod{11}$이므로, 원소 5의 위수는 5이다. 이는 전체 군의 위수 10과 일치하지 않으므로 5는 생성자가 아니다.

Cyclic Groups and Generators

- Setting: 곱셈군 Z_{11}^{*}, 군의 위수 $m = 10$.
  - Case 2: $g = 2$, 원소 2를 거듭제곱하여 모듈러 11 연산을 수행.

Cyclic Groups and Generators

- $2^1 \equiv 2 \pmod{11}$
- $2^2 \equiv 4 \pmod{11}$
- $2^3 \equiv 8 \pmod{11}$
- $2^4 \equiv 16 \equiv 5 \pmod{11}$
- $2^5 = 2^4 \times 2 \equiv 5 \times 2 = 10 \pmod{11}$
- $2^6 = 2^5 \times 2 \equiv 10 \times 2 = 20 \equiv 9 \pmod{11}$
- $2^7 = 2^6 \times 2 \equiv 9 \times 2 = 18 \equiv 7 \pmod{11}$
- $2^8 = 2^7 \times 2 \equiv 7 \times 2 = 14 \equiv 3 \pmod{11}$
- $2^9 = 2^8 \times 2 \equiv 3 \times 2 = 6 \pmod{11}$
- $2^{10} = 2^9 \times 2 \equiv 6 \times 2 = 12 \equiv 1 \pmod{11}$

- 결론: 부분군 ⟨2⟩ = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}이 되어 위수 10과 크기가 동일하다.
  따라서 2는 Z_{11}^*의 생성자이다.